Assignment-02

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Cryptography for Network Security - CSA511

1. DES problem : Given a plaintext block and a simplified DES key, perform one round of DES and show all intermediate value.

98/100

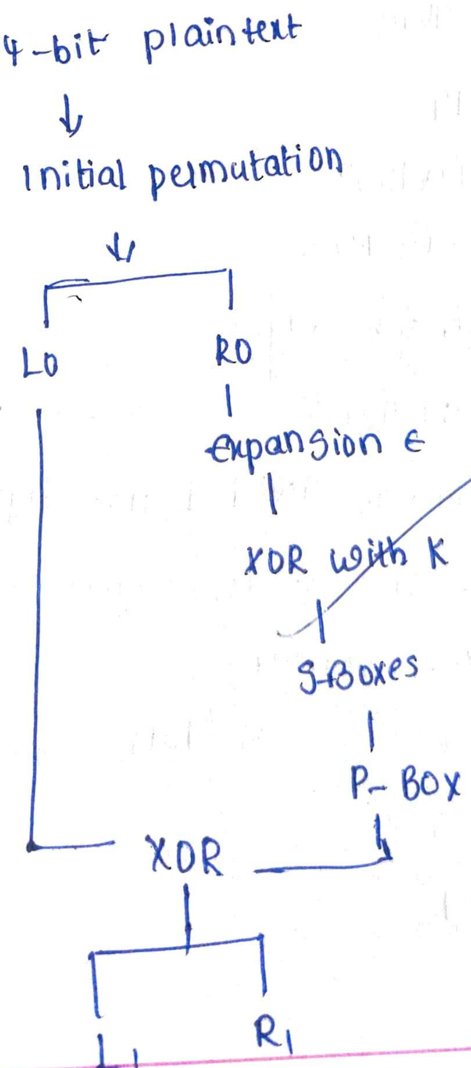
Given

Plain text
P = 123456 ABCD 132536

key: K = AABB0918 2736CCDD

DES operates on a 64-bit plaintext block and uses a 64-bit key (with 56 effective key bits).

DES Structure



Step1: Initial permutation

After applying the DES initial permutation:

IP (P) = CC00CCFFF0AAF0AA

∵ L_0 = CC00CCFF

R_0 = F0AAF0AA

Step 2: Expansion

The right half is 32 bits.

DES expands it from 32 bits to 48 bits

R_0 = F0AAF0AA

After expansion:

$E(R_0)$ = 7A15557A1555

Step 3: XOR with Round key

k_1 = 1B02EFFC7072

Then

$E(R_0) \oplus K_1$

= 7A15557A1555

$\oplus$ 1B02EFFC7072

= $\boxed{6117BA866527}$

Step4: S-Box Substitution

The 48 bit result is divided into eight groups of 6 bit

Each group enters one of the eight DES S-boxes

The eight S-boxes convert:

48 bits → 32 bits

Result:

5C82B597

Step 5: Box permutation

The 32-bit S-box output is rearranged using the DES P-box

$P(5C82B597) = 234AA9BB$

Step 6: Calculate R_1

The DES round function is:

$R_1 = L_0 \oplus F(R_0, K_1)$

$\therefore R_1 = CC00CCFF \oplus 234AA9BB$

$R_1 = EF4A6544$

and $L_1 = R_0$

$\boxed{L_1 = F0AAF0AA}$

2. Compare Symmetric and Asymmetric encryption using real time applications (eg. whats App vs email encryption).

Symmetric vs Asymmetric encryption

feature	symmetric encryption	Asymmetric encryption
Keys	one shared key	public + private key
speed	very fast	slower
key distribution	Difficult	Easier
Encryption / Decryption	same secret key	Different keys

Examples	AES, DES Chacha20	public/private key RSA, ECC
Best use	Large amount of data	key exchange, digital signatures

Real-time application - whatsApp

WhatsApp uses end-to-end encryption, primarily based on the signal protocol.

A simplified process is:

```
Alice                         Bob
  |                            |
  |--- key establishment ---->|
  |                            |
  |=== encrypted message ===>|
  |                            |
```

Symmetric cryptography is used for efficiently encrypting the actual conversation data, while public-key techniques help.

establish / authenticate the cryptographic keys

Email.

Email systems can use both approaches

for example, PGP / GPG can use:

1. RSA / ECC → securely exchange or protect a key.
2. AES → encrypt the actual email / message.

Message

|

AES encryption

|

Encrypted message

|

AES key protected using
receiver's public key

Conclusion:

Symmetric encryption is faster and suitable for large data.

- Asymmetric encryption solves the key distribution problem and is useful for authentication, key exchange, and digital signatures.

3. RSA problem choose two small primes $p=7, q=11$. Generate RSA keys and encrypt the message $M=9$.

Given:

$p=7, \quad q=11$

message: $M=9$

Step 1: Calculate n

$n = pq$

$n = 7 \times 11 = 77$

Step 2: Calculate Euler's Totient

$$\phi(n) = (p-1)(q-1)$$

$$\phi(77) = 6 \times 10 = 60$$

Step 3: Choose public exponent e

we need:

$$1 < e < 60$$

and

$$\gcd(e, 60) = 1$$

choose:

$$\boxed{e = 7}$$

since:

$$\gcd(7, 60) = 1$$

the choice is valid.

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Step 4: calculate private key d

we'd need:

$ed \equiv 1 \pmod{60}$

$\therefore 7d \equiv 1 \pmod{60}$

since:

$7 \times 43 = 301$

and $301 \bmod 60 = 1$

we get:

$\boxed{d = 43}$

RSA keys

public key: $(e, n) = (7, 77)$

private key: $(d, n) = (43, 77)$

Step 5: Encryption

RSA encryption

$C = M^e \bmod n$

$C = 9^7 \bmod 77$

Calculate: $9^2 = 81 \equiv 4 \pmod{77}$

$9^4 = 16 \pmod{77}$

$\because 9^7 = 9^4 \times 9^2 \times 9$

$= 16 \times 4 \times 9$

$= 576$

$576 \bmod 77 = 37$

4. Explain CBC and CTR modes of operation of with block diagrams. which one is more secure for parallel processing?

Given

CBC - cipher Block Chaining

In CBC, each plaintext block is XORed with the previous Ciphertext block before encryption.

for the first block, an initalization vector (IV) is used.

Formula

$$C_1 = E_K(P_1 \oplus IV)$$

$$C_i = E_K(P_i \oplus C_{i-1})$$

Block diagram

IV
↓
P1 --- XOR ---> Encryption
↓
C1
↓
--- XOR ---> Encryption
↓
C2
↓
--- XOR ---> Encryption
↓
C3

- features of CBC

  - Requires an IV
  - Each block depends on the previous ciphertext block.
  - Encryption can't be fully parallelized
  - provides confidentiality but does not by itself provide authentication / integrity.
  - padding is generally required for block-aligned encryption.

B. CTR - Counter mode.

CTR converts a block cipher into a stream-like cipher. A counter is encrypted, and the resulting keystream is XORed with plaintext.

<u>formula</u>

$$C_i = P_i \oplus E_K(CTR_i)$$

<u>Block Diagram</u>

Counter 1 → Encryption → Keystream 1

P_1 → XOR → C_1

counter 2 → Encryption → keystream 2

P_2 → XOR → C_2

Counter 3 → Encryption → Keystream 3

P_3 → XOR → C_3

Features of CTR

- No padding is required for arbitrary length data
- Encryption blocks can be processed independently
- Very suitable for parallel processing.
- Encryption and decryption use the same operation
- The counter/nonce combination must never repeat with the same key.

CBC vs CTR for parallel processing.

Feature	CBC	CTR
Encryption parallelism	No	Yes
Decryption	partly possible	yes
padding	usually required	Not required
Random access	poor	Good
performance	lower for parallel systems	Higher

Answer: CTR is better for parallel processing.

5. Define quantum cryptography. How does Quantum Key Distribution (QKD) solve the key distribution problem

Definitions

Quantum cryptography is a cryptographic approach that uses principles of quantum mechanics to provide secure communication.

The most important application is:

$\boxed{\text{Quantum key Distribution (QKD)}}$

QKD allows two parties to establish a shared secret key while detecting whether someone has tried to intercept the communication.

How QKD solves the key - distribution problem.

In traditional cryptography, Alice needs to securely provide a secret key to Bob.

The problem is:

Alice --- secret key ---> Bob

↑

eve?

If an attacker intercepts the key security is compromised.

QKD approach:

QKD uses quantum state, commonly photons, to transmit information.

A famous QKD protocol is BB 84

Simplified process:

Alice
↓ Quantum state
Quantum channel
↓
Bob
↓ --- public discussion ---
↓
Key verification
↓
Shared secret key

Why is interception detected?

One fundamental principle is that measuring an unknown quantum state can disturb it.

∴ Alice -- Quantum state

Alice --- Quantum state → bob
↑
Eve
|
measurement
|
Disturbance.